

Cheng-Yan(Jay), Tsai

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WORK EXPERIENCES

AES Mega Co., Ltd.

Taipei, Taiwan

SOLIDWORKS PREMIUM, SOLIDWORKS ELECTRICAL, AUTOCAD, PLC, MATLAB

Assistant Manager

Jan 2025-present

- Served as design lead for project-level modeling and verification activities.
- Developed and delivered 10 internal technical documents to train engineers on SOLIDWORKS usage and design work-flow standards.
- Led automation system optimization for Hu-Shan project, focusing on electromechanical integration and control system commissioning, reducing manual operation by 80% and improving start-up time by 1.5 hours.

Electromechanical Engineer

Aug 2023-Dec 2024

- Designed and optimized electromechanical subsystems, developing system layouts, and generating bills of materials for Hu-Shan(1.5MW) hydropower plant and Liu-Huang-Zi-Ping(4.2MW) geothermal power plant projects.
- Executed control software optimization for Li-Cha(130kW) hydropower plant project, improving operational stability.

Pei-I Learning Center

Taipei, Taiwan

Electrical Machinery Teacher

June 2022-June 2025

EDUCATION

McGill University

Montréal, QC, Canada

M.Eng in Electrical and Computer Engineering

Jan 2026-Dec 2027 (Expected date)

National Taiwan Normal University (NTNU)

Taipei, Taiwan

B.S. in Vehicle and Energy Engineering

Sept 2019-June 2023

PROJECTS

Building Energy Efficiency Laboratory

NTNU

Experimenter

Sept 2022-June 2023

- Designed a desiccant wheel and optimized its efficiency by 30% through adjusting the proportion of regeneration, process, and purge sector.
- Awarded the Excellent Award in a research competition.

Omron Taiwan Electronics Inc.

Taipei, Taiwan

PLC Trainee

Aug 2021

- Practiced advanced skills of PLC ladder programming, appliances of human-machine interfaces, connection between internet and devices.

Gogoro Taiwan Limited.

Taipei, Taiwan

Trainee

Jan 2022